

A New Family of Normal Numbers

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Abstract

We study the numbers $\xi_{b,c;a}$, obtained by restricting the logarithmic series to products of powers of a and c . For integers $a, b, c \geq 2$ with $\gcd(c, ab) = 1$, we prove normality in base b . More strongly, fixing a, b and varying the eligible parameter c , every finite rational combination of distinct members with coefficients not all zero is normal, even after a rational translation. The construction extends Stoneham numbers and includes values of Dibag's localized logarithms. The proof follows the denominators of rational approximations: one prime detects a contribution that the others cannot cancel. Both main theorems have complete Lean formalizations.

1 Meet the Xi family

A real number is *normal in base b* if every word of ℓ base- b digits occurs with limiting frequency $b^{-\ell}$, counting overlapping occurrences. Thus a binary normal number contains each digit half the time, each two-digit word a quarter of the time, and so on. For numbers outside $[0, 1)$, we use the expansion of the fractional part.

The numbers in this paper have a simple analytic definition. For positive integers a, c , write

$$\mathcal{M}(a, c) = \{a^i c^k : i, k \geq 0\},$$

with each integer included only once. For an integer $b \geq 2$, define

$$\xi_{b,c;a} = \sum_{m \in \mathcal{M}(a,c)} \frac{1}{mb^m}. \quad (1)$$

We call these numbers the *Xi family*. The first index is the expansion base; the index after the semicolon is an additional multiplicative generator.

The series converges, since it is a subseries of

$$\sum_{m \geq 1} \frac{1}{mb^m} = -\log(1 - 1/b).$$

When $a, c \geq 2$ are coprime, every product $a^i c^k$ has a unique representation, so (1) becomes the double series

$$\xi_{b,c;a} = \sum_{i,k \geq 0} \frac{1}{a^i c^k b^{a^i c^k}}. \quad (2)$$

Indeed, an equality between two such products would equate a positive power of a with a positive power of c , unless both pairs of exponents agree. Coprimality excludes that possibility.

For example, $\xi_{2,3;2}$ uses the entries of the grid

	1	3	9	27	...	
1	1	3	9	27	...	
2	2	6	18	54	...	
4	4	12	36	108	...	
8	8	24	72	216	...	
⋮	⋮	⋮	⋮	⋮	⋮	

each entry m contributes $1/(m2^m)$.

Its value begins 0.68562264189340237754... The rule selects terms of a familiar convergent series; it makes no reference to digit frequencies.

Theorem 1 (Normality of the Xi family). *Let $a, b, c \geq 2$ be integers with $\gcd(c, ab) = 1$. Then $\xi_{b,c;a}$ is normal in base b .*

The condition allows a to share prime factors with the base. It covers $\xi_{2,3;2}$, for instance, and also the composite-generator example $\xi_{2,35;6}$.

The arithmetic property is stronger. Addition can destroy normality: if $x \in (0, 1)$ is normal, so is $1 - x$, but their sum is an integer. Within the following fixed-parameter Xi family, rational combinations behave differently.

Theorem 2 (Rational combinations). *Fix integers $a, b \geq 2$. Let $c_1, \dots, c_t \geq 2$ be distinct integers with $\gcd(c_j, ab) = 1$, and let $q_0, \lambda_1, \dots, \lambda_t \in \mathbb{Q}$. If at least one λ_j is nonzero, then*

$$q_0 + \lambda_1 \xi_{b,c_1;a} + \dots + \lambda_t \xi_{b,c_t;a} \quad (3)$$

is normal in base b .

The parameters may share primes or be powers of one another. In particular, the binary-normal examples include

$$\xi_{2,3;2} - \xi_{2,9;2} \quad \text{and} \quad \xi_{2,3;2} - 2\xi_{2,9;2} + \frac{1}{7}\xi_{2,15;2}.$$

The first example removes every term whose exponent of three is even. An infinite portion of the series cancels, and the remainder is still normal. Accounting for such cancellations is the main work of the proof.

Stoneham numbers and localized logarithms

The usual Stoneham constant is

$$\alpha_{b,c} = \sum_{k \geq 1} \frac{1}{c^k b^{c^k}}.$$

Stoneham proved foundational normality results for these series; Bailey and Crandall established base- b normality for every coprime pair $b, c \geq 2$ [1, 2]. Setting the additional generator to one in our distinct-index definition gives

$$\xi_{b,c;1} = \frac{1}{b} + \alpha_{b,c}. \quad (4)$$

Thus the construction passes from one geometric sequence of exponents to a grid of products of powers. Formula (4) uses (1): substituting $a = 1$ in the double series would repeat every term infinitely. The boundary $a = 1$ is the classical Stoneham case; our two theorems concern $a \geq 2$.

There is also an older analytic origin. Dibag's localized logarithm retains the terms of $\log(1 + z)$ whose exponents have prime factors in a chosen set [3, Definitions 1.1–1.2]. For distinct prime generators a, c , our function

$$F_{a,c}(z) = \sum_{m \in \mathcal{M}(a,c)} \frac{z^m}{m}$$

is precisely $-L_{\{a,c\}}(-z)$ in Dibag's notation, and $\xi_{b,c;a} = F_{a,c}(1/b)$. Composite generators allow further restrictions on the exponents. The results here establish normality and rational-combination stability for this family.

Our argument belongs to the arithmetic approach to normal numbers developed by Stoneham, Korobov, Kano, Bailey and Crandall, and others. Earlier rings of normal numbers provide precedents for arithmetic closure [4, 5]. We use a rational-orbit estimate of Vandehey [6]; the accompanying Lean proof independently establishes the estimates it needs by repeated differencing.

2 Read the digits through rational approximations

Multiplication by b^n moves the digit at position $n + 1$ to the front. A standard form of Weyl's criterion says that x is normal in base b exactly when

$$\sum_{n < N} e(hb^n x) = o(N) \quad \text{for every integer } h \neq 0, \quad e(u) = \exp(2\pi i u). \quad (5)$$

Indeed, uniform distribution of $\{b^n x\}$ gives each interval $[j/b^\ell, (j+1)/b^\ell)$ its length as frequency. These are exactly the intervals encoding digit words [7].

Here is the analytic input. For fixed b , a finite set P of primes not dividing b , and any $\varepsilon > 0$, there are constants $C, \delta > 0$ such that

$$\left| \sum_{r=0}^{L-1} e(Ab^r/q) \right| \leq CLq^{-\delta}(1 + \log q), \quad L \geq q^\varepsilon, \quad (6)$$

whenever A/q is reduced and every prime factor of q lies in P . This is Vandehey's Corollary 1.2 [6], with the starting index shifted. The estimate says that a sufficiently long piece of a rational radix orbit becomes equidistributed as its denominator grows. All logarithms below are natural.

We will use the following elementary consequence. It is stated for weighted series because subtracting Xi numbers naturally introduces weights.

Proposition 1 (A denominator test). *Fix $a, b \geq 2$ and integers $d_1, \dots, d_g \geq 2$ coprime to ab . Let each $w_j(k)$ be a bounded rational sequence, with a common denominator for all its values. Put*

$$x = q_0 + \sum_{j=1}^g \sum_{i, k \geq 0} \frac{w_j(k)}{a^i d_j^k b^{a^i d_j^k}}, \quad (7)$$

$$T_n = q_0 + \sum_{j=1}^g \sum_{a^i d_j^k \leq n} \frac{w_j(k)}{a^i d_j^k b^{a^i d_j^k}}, \quad q_0 \in \mathbb{Q}. \quad (8)$$

Suppose that, for some prime $p \nmid b$ and some $\eta > 0$, the reduced denominator of T_n contains a factor at least n^η that is a power of p , except at a set of positions of density zero. Then x is normal in base b .

A set has *density zero* if its intersection with $\{1, \dots, N\}$ has $o(N)$ elements. Thus occasional denominator cancellation is allowed; it must occupy a vanishing fraction of digit positions.

Proof. Set $R_n = b^n T_n$. Boundedness of the weights and uniqueness of the products within each group give

$$|b^n x - R_n| \ll \sum_{m > n} \frac{b^{n-m}}{m} \leq \frac{1}{(b-1)(n+1)}. \quad (9)$$

Here $F \ll G$ means $|F| \leq C_0 G$ for a constant depending only on the fixed data. All implied constants in this proof have this meaning, with additional dependence on a fixed Fourier frequency allowed.

Let D clear the coefficient denominators, including that of q_0 . Every denominator of R_n divides

$$Da^{\lfloor \log_a n \rfloor} \prod_{j=1}^g d_j^{\lfloor \log_{d_j} n \rfloor} \leq Dn^{g+1}. \quad (10)$$

There are only $O((\log N)^2)$ insertion positions $a^i d_j^k \leq 2N$. Between successive insertions, $R_{n+r} = b^r R_n$ exactly.

Some factors of a or D may divide b . On a dyadic interval $[N, 2N)$, remove each insertion position and the following $C_1 \log(2N)$ positions, including any such intervals crossing N . For sufficiently large fixed C_1 , every denominator prime dividing b has then been cleared by multiplication by b . To see this term by term, its possible denominator exponent at a prime dividing b is $O(\log N)$, whereas each new radix shift supplies a fixed positive power of that prime. The fixed rational term clears too. The removed positions number $O((\log N)^3)$.

The remaining positions form $O((\log N)^2)$ intervals on each of which the reduced denominator is constant, coprime to b , supported on a fixed finite set of primes, and at most DN^{g+1} up to a fixed factor. Our hypothesis removes another $o(N)$ positions: discard the intervals whose p -part is smaller than N^η . Each position in such an interval is exceptional, because $p \nmid b$ and the p -part is constant there.

Fix $h \neq 0$. Multiplication by h reduces a denominator by at most $|h|$. Thus on each retained interval, the denominator q of hR_n satisfies

$$q \gg_h N^\eta, \quad q \ll N^{g+1}.$$

Discard intervals shorter than $N^{1/2}$; their total length is $O(N^{1/2}(\log N)^2) = o(N)$. Choose $0 < \varepsilon < 1/(2(g+1))$. Every remaining interval has length at least q^ε for large N , so (6) bounds its Fourier sum by its length times $O_h(N^{-\eta\delta} \log N)$. Adding the intervals gives

$$\sum_{N \leq n < M} e(hR_n) = o(N), \quad N < M \leq 2N,$$

uniformly in M ; a final truncated interval is handled by the same short-interval rule. By (9) and $|e(u) - e(v)| \leq 2\pi|u - v|$, replacing R_n by $b^n x$ costs only $O_h(1)$ on this scale. Summing dyadic intervals proves (5) through every prefix length. $\square$

The analytic task is now complete. To prove normality, it remains to find one prime whose denominator powers survive most of the time.

3 Why a denominator survives

For a prime p , let $v_p(y)$ be its exponent in a nonzero rational number y , and put $v_p(0) = +\infty$. A negative valuation records a denominator: for example, $v_3(2/27) = -3$. We will repeatedly use the elementary rule

$$v_p(u + v) = \min(v_p(u), v_p(v)) \quad \text{if } v_p(u) \neq v_p(v). \quad (11)$$

More generally, a finite sum with a unique term of smallest valuation has that same valuation.

Fix $a, b, d \geq 2$ with $\gcd(d, ab) = 1$, a nonzero periodic rational sequence $w(k)$, and a prime $p \mid d$. Write

$$S_n = \sum_{a^i d^k \leq n} \frac{w(k)}{a^i d^k b^{a^i d^k}}, \quad D_p(n) = \max\{0, -v_p(S_n)\}, \quad (12)$$

where $D_p(n) = 0$ when $S_n = 0$. Thus $D_p(n)$ is the exponent of p in the reduced denominator of the whole truncation.

Lemma 1 (The denominator slope). *With the preceding notation,*

$$\frac{D_p(n)}{\log n} \longrightarrow \sigma_p(d) := \frac{v_p(d)}{\log d} \quad \text{in density.} \quad (13)$$

That is, for every positive tolerance, the positions at which the two sides differ by more than that tolerance have density zero.

The slope has a simple meaning. Among the terms already inserted by position n , the largest possible exponent of d is about $\log_d n$. It contributes about $v_p(d) \log_d n$ powers of p to the denominator. The lemma says that cancellation changes this leading size only rarely.

We first record the spacing fact that makes the argument work.

Lemma 2 (Small relative gaps). *If $u, v \geq 2$ have no equal positive powers, then for every $\varepsilon > 0$, every sufficiently large interval $(x, (1 + \varepsilon)x]$ contains a number $u^i v^k$ with $i, k \geq 0$.*

Proof. The ratio $\log u / \log v$ is irrational. Its integer multiples are dense modulo one. Choose finitely many nonnegative multiples whose fractional parts have gaps smaller than $\log(1 + \varepsilon) / \log v$. For any sufficiently large $\log x$, one of the corresponding values $i \log u$, followed by a nonnegative multiple of $\log v$, lies in $(\log x, \log x + \log(1 + \varepsilon)]$. Exponentiate. $\square$

Proof of Lemma 1. Put $v = v_p(d)$ and $\sigma = v / \log d$. Every nonzero summand at position $a^i d^k$ has valuation

$$v_p(w(k)) - kv, \quad (14)$$

since $p \nmid ab$. There are only finitely many nonzero weights, so $k \leq \log_d n$ immediately gives

$$D_p(n) \leq \sigma \log n + O(1).$$

The lower bound requires three observations.

Equal valuations cannot occur too close together. If two expressions in (14) agree, the difference of their k -coordinates belongs to a fixed finite set. The ratio of the two insertion positions is therefore $a^r d^s$, where $r \in \mathbb{Z}$ and s lies in that finite set. For each fixed s , the logarithms $r \log a + s \log d$ form a discrete set. After excluding zero, their distances from zero have a positive lower bound, uniformly over these finitely many s .

Choose $\Lambda > 1$ smaller than the resulting separation ratio. Distinct nonzero terms inserted in any interval $[X, \Lambda X]$ then have distinct valuations. Any nonempty sum of such terms has the valuation of its unique smallest-valuation term.

Deep denominator terms occur frequently on a relative scale. Fix $0 < \theta < \sigma$. Let $L \geq 1$ be a period of w , and choose k_0 with $w(k_0) \neq 0$. At scale N , choose the least nonnegative $k_N \equiv k_0 \pmod{L}$ for which

$$k_N v - v_p(w(k_0)) > \theta \log N.$$

Then

$$d^{k_N} \asymp N^{\theta/\sigma}. \quad (15)$$

Every position in the set

$$d^{k_N} \mathcal{M}(a, d^L) \tag{16}$$

carries a nonzero term of valuation smaller than $-\theta \log N$. Here $F \asymp G$ means that their ratio is bounded above and below by positive fixed constants.

The generators a, d^L are coprime and hence have no equal positive powers. By Lemma 2, the set in (16) meets every interval $(x, (1+\varepsilon)x]$, uniformly for $N \leq x \leq 2N$, once N is sufficiently large. Indeed, after division by d^{k_N} , the starting points are at least a constant times $N^{1-\theta/\sigma}$, which tends to infinity.

Extensive cancellation is confined to short intervals. Call a position $n \in [N, 2N]$ bad if $D_p(n) \leq \theta \log N$, equivalently $v_p(S_n) \geq -\theta \log N$. Take two bad positions $x < y$ within a block of multiplicative ratio at most Λ . Their difference $S_y - S_x$ cannot contain an insertion from (16): the inserted terms have distinct valuations, so such an insertion would force $v_p(S_y - S_x) < -\theta \log N$, contradicting the bound at both endpoints.

But a deep insertion occurs before $(1+\varepsilon)x$. Consequently two bad positions in this block are separated by less than $\varepsilon x \leq 2\varepsilon N$. All bad positions in the block therefore lie in an interval of length at most $2\varepsilon N$. Cover $[N, 2N]$ by a fixed finite number of such blocks. Since ε is arbitrary,

$$\#\{n \in [N, 2N] : D_p(n) \leq \theta \log N\} = o(N). \tag{17}$$

This holds for every $\theta < \sigma$. To replace $\log N$ by $\log n$, first increase θ slightly while keeping it below σ ; on this scale the two logarithms differ by at most $\log 2$. Together with the upper bound, and then a dyadic summation, this proves (13). $\square$

4 Normality of each Xi number

We can now prove Theorem 1. Take $d = c$ and $w(k) = 1$ in Lemma 1, and choose any prime $p \mid c$. The hypothesis $\gcd(c, ab) = 1$ gives $p \nmid ab$. Outside a set of density zero, the p -part of the denominator of the truncation is at least

$$p^{\frac{1}{2}\sigma_p(c)\log n} = n^{\frac{1}{2}\sigma_p(c)\log p}.$$

Proposition 1 applies with one group and $q_0 = 0$, proving normality.

The mechanism can be read directly from the series. Between insertions, the shifted truncations follow rational radix orbits. Their denominators become large at almost every position, and the orbit estimate turns that growth into the correct frequency of every digit word.

5 Why rational combinations remain normal

We turn to Theorem 2. Discard zero coefficients. The key is to group parameters that are powers of the same integer *before* comparing their denominators.

First collect the exact overlaps

Every integer $c \geq 2$ has a unique expression $c = d^r$, where $r \geq 1$ and d is not a proper integer power. In its prime factorization, r is the greatest common divisor of the exponents. Two integers have equal positive powers exactly when their corresponding d 's agree.

Group together all parameters with the same d . Their contribution to (3) is

$$Y_d = \sum_{i,k \geq 0} \frac{w_d(k)}{a^i d^k b^{a^i d^k}}, \quad w_d(k) = \sum_{r \in E_d} \lambda_{d,r} \mathbf{1}_{r|k}. \quad (18)$$

Here E_d is the finite set of powers present in the group, and $r \mid 0$ as usual. The sequence w_d is periodic, with period dividing $\text{lcm}(E_d)$.

It is also nonzero. Choose the smallest $r \in E_d$ with nonzero coefficient and evaluate at $k = r$. Any other contributing exponent must be a proper divisor of r , whose coefficient is zero by the choice of r . Hence $w_d(r) = \lambda_{d,r} \neq 0$.

For instance, $\xi_{b,d;a} - \xi_{b,d^2;a}$ has weight one at odd k and zero at even k . Grouping makes this exact cancellation visible, and Lemma 1 applies to the surviving periodic weight.

Then one prime separates the groups

Let $S_{d,n}$ be the truncation of Y_d at insertion position n . Choose a prime p dividing at least one active d . It divides neither a nor b . Associate to each group the slope

$$\sigma_d = \frac{v_p(d)}{\log d},$$

which is zero if $p \nmid d$. Distinct groups cannot have equal positive slopes: equality would give

$$d_1^{v_p(d_2)} = d_2^{v_p(d_1)},$$

forcing the same primitive integer in both groups.

There is therefore a unique group d_* of largest positive slope. Choose $0 < \theta < \sigma_{d_*}$ with $\sigma_d < \theta$ for every other group. By Lemma 1, outside a set of density zero,

$$v_p(S_{d_*,n}) < -\theta \log n.$$

Every other group satisfies the unconditional bound

$$v_p(S_{d,n}) \geq -\sigma_d \log n - O(1).$$

When $p \nmid d$, this simply says that its p -denominator is bounded: only the finitely many rational weights can contribute a factor. The rational constant q_0 has a fixed valuation as well. For all sufficiently large nonexceptional n , the d_* group has strictly smaller valuation than every other contribution. Rule (11) gives

$$v_p\left(q_0 + \sum_d S_{d,n}\right) = v_p(S_{d_*,n}) < -\theta \log n.$$

Thus the p -part of the denominator of the complete truncation exceeds $n^{\theta \log p}$ at a density-one set of positions. Proposition 1 proves that $q_0 + \sum_d Y_d$ is normal in base b , completing the proof of Theorem 2. $\square$

Corollary 1. *For fixed $a, b \geq 2$, the number one and all the values $\xi_{b,c;a}$, with $c \geq 2$ and $\gcd(c, ab) = 1$, are linearly independent over $\mathbb{Q}$. The ξ values span an infinite-dimensional rational vector space whose every nonzero element is normal in base b .*

Proof. A nontrivial rational relation involving a ξ value would make the normal number in Theorem 2 equal to zero. This is impossible: rational numbers have eventually periodic expansions and are not normal. There are infinitely many eligible parameters c . $\square$

The arithmetic theorem fixes both a and b . Within that family it allows every finite rational combination, including negative coefficients and overlapping parameter powers. Its proof identifies what makes this possible: periodic cancellations preserve a definite denominator growth rate, and distinct groups have different rates at a suitable prime.

The accompanying Lean proof

Theorems 1 and 2 are formalized with their stated hypotheses. In the accompanying source package, `XiFamilyChallenge.lean` defines the distinct-index series and normality through overlapping word frequencies. It imports only `mathlib`, independently of the proof development. `XiFamilySolution.lean` proves both statements, and `XiFamilyStatement.lean` establishes correspondence with the definitions used internally.

The formal proof follows the same denominator argument. Its analytic part proves sufficient rational-orbit estimates by repeated differencing, so it does not assume Vandehey’s theorem as an axiom. The complete source build and the endpoint axiom checks passed with Lean 4.34.0-rc2 and pinned `mathlib` dependencies. There are no unfinished proofs or added axioms. Convergence, the double-series identity, and the Stoneham specialization are included in the development.

References

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